$$\sum_{i=1}^{n+1} \frac{1}{i^2+i} = \sum_{i=1}^{n} \frac{1}{i^2+i} + \frac{1}{(n+1)^2+(n+1)}$$

B.2-3

$$\underline{Y} = \underline{X} + N \qquad X \sim N(\mu, R_x), \qquad N \sim N(0, R_n)$$

Find $f(\underline{x}|y)$

$$\underline{Y}|\underline{X} \sim N(\underline{X}, R_n)$$

Likelihood

Bayes Theorem

likelihood

prior probability

$$f(\underline{x}|y) = \frac{f(y|\underline{x})\, f(\underline{x})}{f(y)}$$

$$f(y|x) = k_1 \exp\left[\frac{-1}{2}(y-\underline{x})^T R_n^{-1}(y-\underline{x})\right]$$

$$f(\underline{x}) = k_2 \exp\left[\frac{-1}{2}(\underline{x}-\underline{\mu})^T R_x^{-1}(\underline{x}-\underline{\mu})\right]$$

$$f(y|\underline{x})\, f(\underline{x}) = k_1 k_2 \exp\left[\frac{-1}{2}(y-\underline{x})^T R_n^{-1}(y-\underline{x})\right] \times \exp\left[\frac{-1}{2}(\underline{x}-\underline{\mu})^T R_x^{-1}(\underline{x}-\underline{\mu})\right]$$

C.T.S.

$$= \exp\left[\frac{-1}{2}(x-\mu_{x|y})^T (R_{x|y})^{-1} \underline{x} - \mu_{x|y})\right]$$

Scalar case

$$f(x|y) \sim N(\mu_{x|y}, \sigma^2_{x|y})$$

$$\mu_{x|y} = \frac{\sigma_x^2}{\sigma_x^2+\sigma_n^2}\, y + \frac{\sigma_n^2}{\sigma_x^2+\sigma_n^2}\, \mu_x$$

$$\sigma^2_{x|y} = \frac{\sigma_n^2 \sigma_x^2}{\sigma_x^2+\sigma_n^2}$$

Toeplitz:

Q: We know that for a real Toeplitz matrix, $\boxed{JT^TJ = T \text{ (persymmetric)}}$

Is this still the case for a complex Toeplitz matrix?

$$JT^HJ = T^*$$

$$T_0 = [\, r_0 \,]$$

$$T_1 = \begin{bmatrix} r_0 & r_1 \\ r_1^* & r_0 \end{bmatrix} \qquad T_1 \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} r_1^* \\ r_2^* \end{bmatrix}$$

$$T_2 = \begin{bmatrix} r_0 & r_1 & r_2 \\ r_1^* & r_0 & r_1 \\ r_2^* & r_1^* & r_0 \end{bmatrix} \qquad T_2 \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} r_1^* \\ r_2^* \\ r_3^* \end{bmatrix}$$

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1} \qquad 1 \le k \le n$$

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1} \qquad 1 \le k \le n$$

$$\binom{n+2}{k} = \binom{n+1}{k} + \binom{n+1}{k-1} \qquad 1 \le k \le n+1$$

B.1-1

$$\phi_X(\omega) = \int_{-\infty}^{\infty} f_X(x)\, e^{j\omega x}\, dx = E_X\left[e^{j\omega x}\right]$$

n^{th} moment of X : $E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$

show: $E[X^n] = \dfrac{1}{j^n} \left.\dfrac{d^n \phi_X(\omega)}{d\omega^n}\right|_{\omega=0}$

$$\phi_X(\omega) = E\left[e^{j\omega x}\right]$$

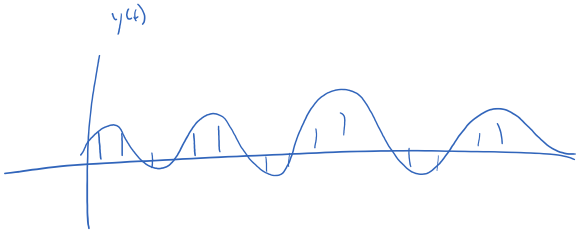
$$= E\left[1 + \frac{(j\omega x)}{1!} + \frac{(j\omega x)^2}{2!} + \frac{(j\omega x)^3}{3!} + \cdots\right]$$

$$= E[1] + E[j\omega x] + \frac{1}{2!} E[(j\omega x)^2] + \frac{1}{3!} E[(j\omega x)^3] + \cdots$$

$$c_x = \iint x\, p(x,y)\, dx\, dy$$

$$c_y = \iint y\, p(x,y)\, dy\, dx$$

$$\int x f(x)\, dx$$



$y[t_1]$

$y[t_2]$

$\vdots$

$y[t_N]$

$$y(t) = A\cos(\omega_1 t + \theta_1) + B\cos(\omega_2 t + \theta_2)$$

known,

want: A, B, θ_1, θ_2.

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$